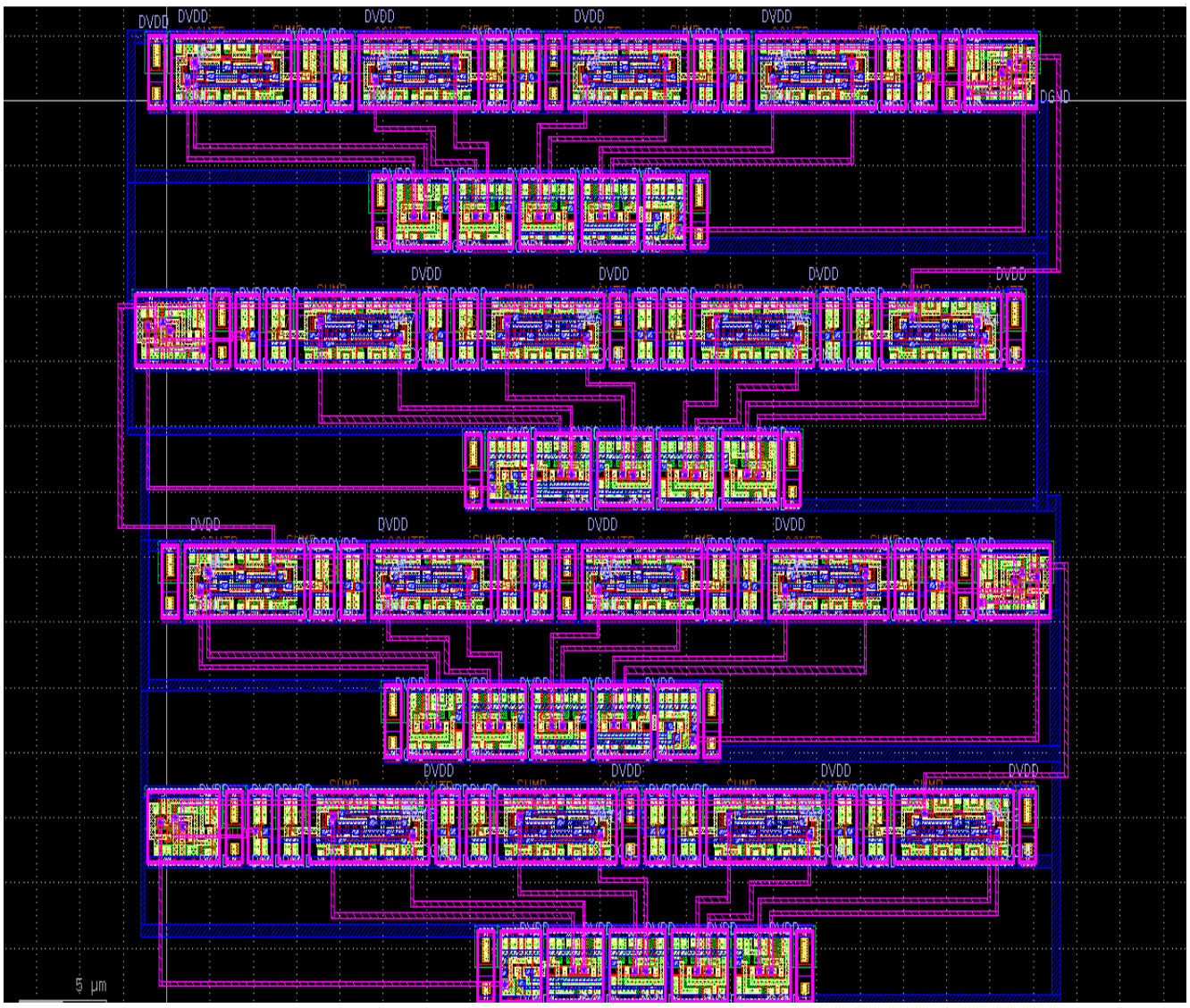
EE5311: Digital IC Design
Tutorial 8

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Roll Number: EE22B032

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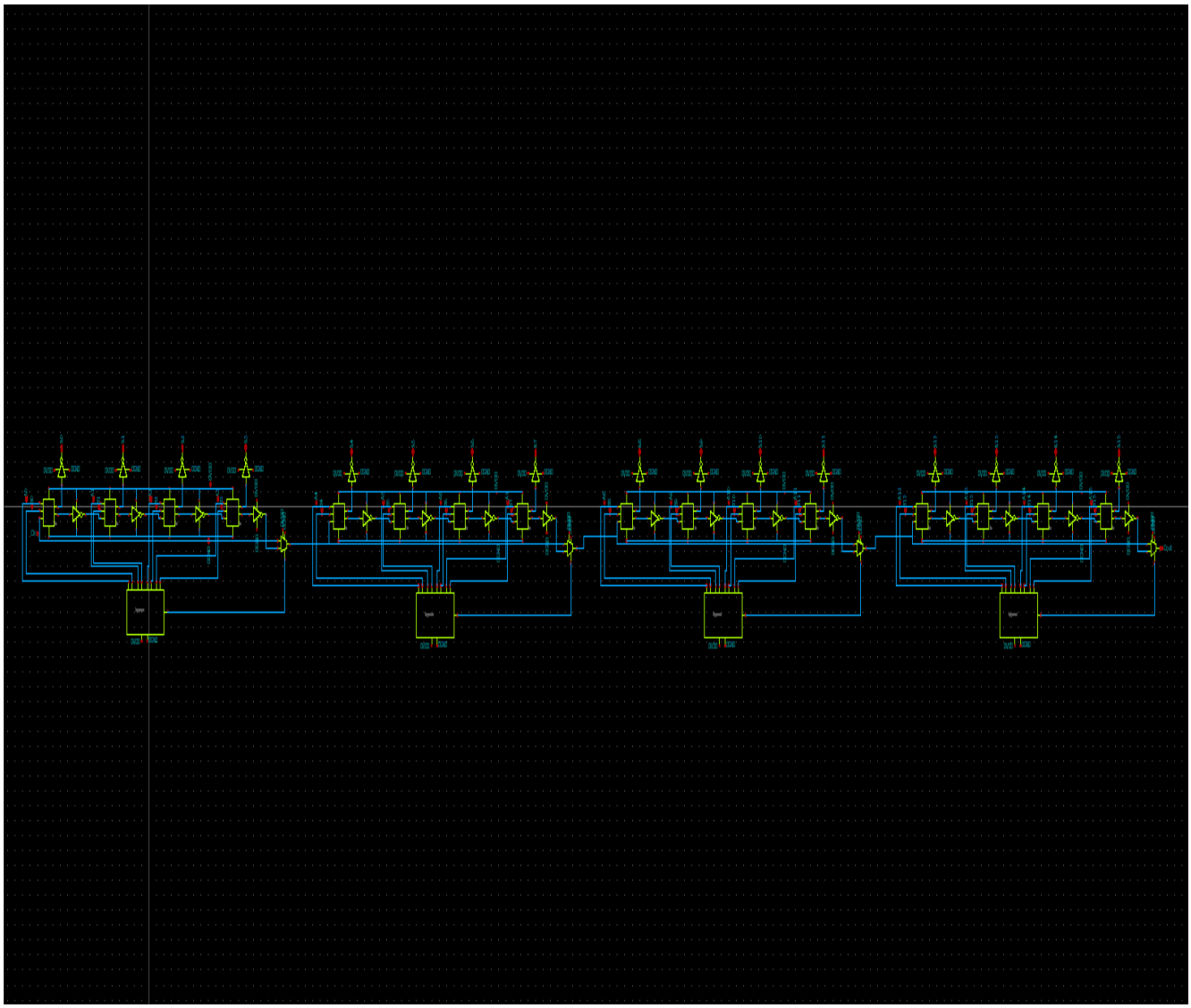
Carry Bypass Adder

Layout



My layout is located at: `/home/EE22B032/ee5311/tutorial_8/cba.gds`

Schematic



Critical Path

Given Inputs:

$$A = \text{FFFF}_{16}, \quad B_0 = 1, \quad B_{1\text{–}15} = 0, \quad Cin = 0$$

Critical Carry Path:

$$B_0 \rightarrow fa0 \rightarrow C_1$$

$$C_1 \rightarrow fa1 \rightarrow C_2$$

$$C_2 \rightarrow fa2 \rightarrow C_3$$

$$C_3 \rightarrow fa3 \rightarrow C_4 \quad \text{(End of Block 0 ripple)}$$

$$C_4 \rightarrow \text{Bypass0 MUX} \rightarrow C_8 \quad \text{(Block 1 propagates)}$$

$$C_8 \rightarrow \text{Bypass1 MUX} \rightarrow C_{12} \quad \text{(Block 2 propagates)}$$

$$C_{12} \rightarrow fa12 \rightarrow C_{13}$$

$$C_{13} \rightarrow fa13 \rightarrow C_{14}$$

$$C_{14} \rightarrow fa14 \rightarrow C_{15}$$

$$C_{15} \rightarrow fa15 \rightarrow Cout \quad \text{(Final Block ripple)}$$

Outputs Affected: $S_{12}, S_{13}, S_{14}, S_{15}, Cout$

Reason (Brief):

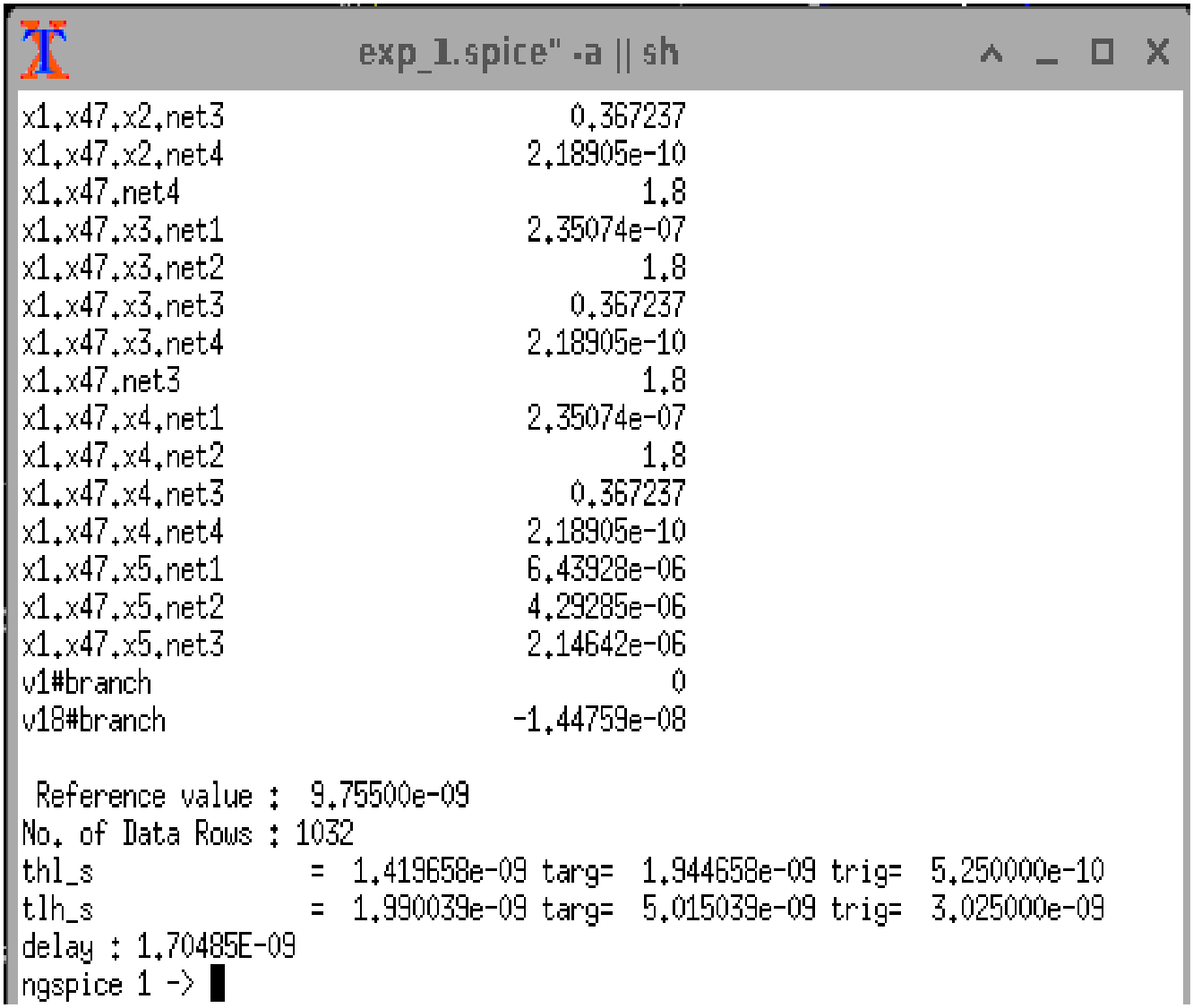
- Bit 0 generates a carry because $1 + 1 = 10_2$, so $G_0 = 1$.
- All upper bits have $P_i = 1$, so the carry bypasses Blocks 1 and 2 through the bypass multiplexers.
- In the final block, the carry must ripple through $fa12$ to $fa15$ to reach $Cout$.

Since this path includes:

$$\text{Carry Generation} \rightarrow \text{Bypass Across Blocks} \rightarrow \text{Final Ripple}$$

it forms the **longest dependent delay**, making it the **critical path**.

Delay without Parasitics



Delay with Parasitics

